



ISO/TC 12 "Quantities and units"
Secretariat: SIS
Committee Manager: Radhe Sven Mr



Secretariats report 2021

Document type	Related content	Document date	Expected action
Meeting / Secretariat report	Meeting: VIRTUAL 26 May 2021	2021-03-16	INFO

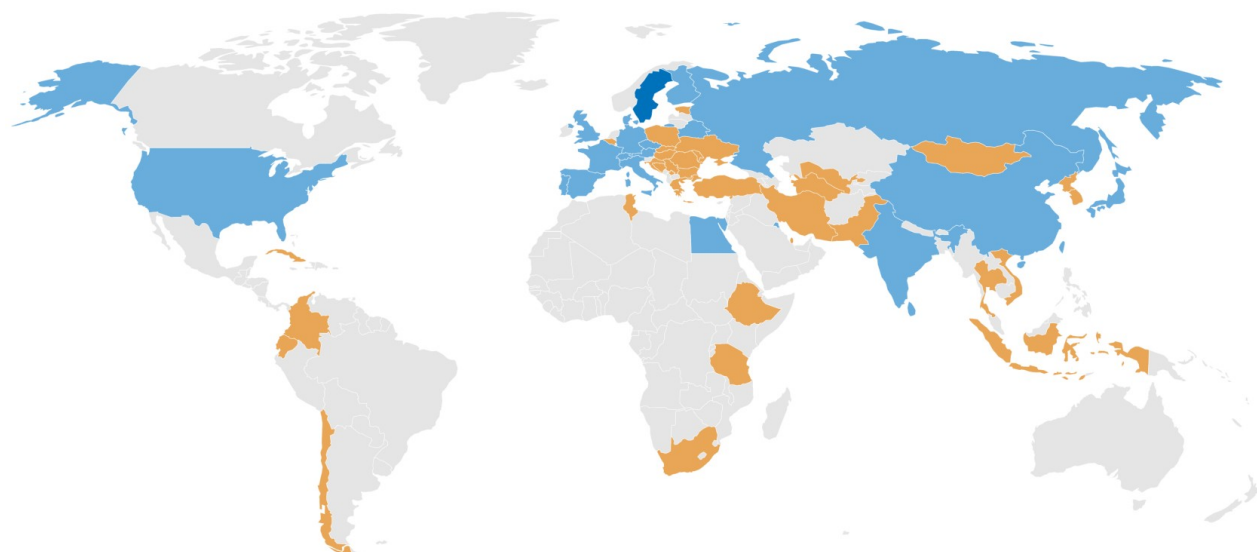
Description

Report of the Secretariat to the meeting of ISO/TC 12 May 26, 2021

ISO/TC 12 Quantities and units

Report of the Secretariat to the meeting of ISO/TC 12 to be held virtual via Zoom, May 26, 2021.

Year of creation	1947
Secretariat	SIS, Sweden Since 1982 the secretariat is held by SIS.
Chairperson	Prof Leslie Pendrill (until end 2021)
Secretary	Mr Sven Radhe, SIS
Secretariat support	Ms Anette Lindor Norén, SIS
ISO Technical Programme Manager	Ms Mercè Ferrés Hernandez
ISO Editorial Programme Manager	Ms Mercè Ferrés Hernández
Scope of ISO/TC 12	Standardization of units and symbols for quantities and units (and mathematical symbols) used within the different fields of science and technology, giving, where necessary, definitions of these quantities and units. Standard conversion factors between the various units.



● Participating Countries (20)

Austria (ASI)
Belarus (BELST)
China (SAC)
Czech Republic (UNMZ)
Denmark (DS)
Egypt (EOS)
Finland (SFS)
France (AFNOR)
Germany (DIN)
India (BIS)
Italy (UNI)
Japan (JISC)
Kuwait (KOWSMD)
Portugal (IPQ)
Russian Federation (GOST R)
Spain (UNE)
Sweden (SIS)
Switzerland (SNV)
United Kingdom (BSI)
United States (ANSI)

● Observing Countries (36)

Belgium (NBN)
Bosnia and Herzegovina (BAS)
Bulgaria (BDS)
Chile (INN)
Colombia (ICONTEC)
Croatia (HZN)
Cuba (NC)
Ecuador (INEN)
Estonia (EVS)
Ethiopia (ESA)
Greece (NQIS ELOT)
Hong Kong (ITCHKSAR)
Hungary (MSZT)
Indonesia (BSN)
Iran, Islamic Republic of (ISIRI)
Korea, Democratic People's Republic of (CSK)
Korea, Republic of (KATS)
Mauritius (MSB)
Moldova, Republic of (ISM)
Mongolia (MASM)
Pakistan (PSQCA)
Poland (PKN)
Qatar (QS)
Romania (ASRO)
Serbia (ISS)
Slovakia (SOSMT)
South Africa (SABS)
Sri Lanka (SLSI)
Tanzania, United Republic of (TBS)
Thailand (TISI)
Tunisia (INNORPI)
Turkey (TSE)
Turkmenistan (MSST)
Ukraine (DSTU)
Uzbekistan (UZSTANDARD)
Viet Nam (STAMEQ)

SLSI (Sri Lanka) moved from P-member to O-member 2020.

NOTE *P-members* (participating members) participate actively in the work, with an obligation to vote on all questions formally submitted for voting within the TC or SC, on new work item proposals, enquiry drafts and final draft International Standards, and to contribute to meetings. *O-members* (observers) follow the work as an observer, and therefore receive committee documents and have the right to submit comments and to attend meetings.

Liaison Committees from ISO/TC 12

ISO/TC 12 can access the documents of the committees below:

Reference	Title	ISO/ IEC	Liaison representative
CIE	International Commission on Illumination	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/IEC JTC 1/SC 32	Data management and interchange	ISO/IEC	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 8/SC 6	Navigation and ship operations	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 37	Language and terminology	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43	Acoustics	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 43/SC 1	Noise	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 47	Chemistry	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 69	Applications of statistical methods	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 85	Nuclear energy, nuclear technologies, and radiological protection	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 115	Pumps	ISO	Ms. Lindor Norén, Anette Mr. Radhe, Sven
ISO/TC 215	Health informatics	ISO	Dr. Cullati, Stéphane Prof. Pendrill, Leslie

IEC/TC 1	Terminology	IEC	Mr. Radhe, Sven
IEC/TC 25	Quantities and units	IEC	Dr. Ehrlich, Charles D

Liaison Committee to ISO/TC 12

The committees and organisations below can access the documents of ISO/TC 12:

Reference	Title	ISO/IEC	Liaison representative
ISO/IEC JTC 1/SC 32	Data management and interchange	ISO/IEC	Ms. Miller, Michaela Ms Rajchel, Lisa Mr Wennblom, Phil
ISO/TC 37	Language and terminology	ISO	Mr Cheng, Yonghong Dr Galinski, Christian Mr Wang, Haitao Mr Zhou, Changqing
ISO/TC 43	Acoustics	ISO	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 1	Noise	ISO	Mr Kornprobst, Hans-Joachim Ms Kursch, Stephanie Ms Liebing, Daniela
ISO/TC 43/SC 3	Underwater acoustics	ISO	Mr Ainslie, Michael
ISO/TC 47	Chemistry	ISO	Ms Nonaka, Reiko
ISO/TC 69	Applications of statistical methods	ISO	Ms Admussen, Jennifer
ISO/TC 85	Nuclear energy, nuclear technologies, and radiological protection	ISO	Mr Balcaen, Eric
ISO/TC 215	Health informatics	ISO	Dr Geßner, Christof Ms Stanford, Sharon Ms Warner, Diana
IEC/TC 25	Quantities and units	IEC	Prof Mari, Luca

Organizations in liaison (Category A and B)

ISO/TC 12 has liaisons with the following organisations and :

Acronym	Title	Category	Liaison representative
BIPM	International Bureau of Weights and Measures	A	Dr Milton, Martin Mr Sitton, Robert
CIE	International Commission on Illumination	A	Ms Nield, Kathryn Dr Zwick, Peter
EC - European Commission	European Commission	B	
EURAMET	European Association of National Metrology Institutes	A	Mr Jarvis, Duncan
IAEA	International Atomic Energy Agency	A	
ICAO	International Civil Aviation Organization	A	
ICRU	International Commission on Radiation Units and Measurements	A	Dr Schauer, David A.
IMO	International Maritime Organization	B	Ms Allnutt, Sandra Rita
Infoterm	International Information Centre for Terminology (Infoterm)	A	Dr Galinski, Christian
ITU	International Telecommunication Union	A	Mr Caras, John George Ms Xiaoya Yang, xiaoya (ITU-T SG 17)
IUCr	International Union of	A	Prof Brock, C.P.

	Crystallography		Mr Dacombe, Michael
IUPAC	International Union of Pure and Applied Chemistry	A	Prof Mills, Ian
IUPAP	International Union of Pure and Applied Physics	A	Dr Karshenboim, Savely Prof Pendrill, Leslie
OASIS	Organization for the Advancement of Structured Information Standards (OASIS)	A	Mr Olken, Frank
OIML	International Organization of Legal Metrology	A	Mr Goblirsch, Richard
UNECE	United Nations Economic Commission for Europe	A	Mr Henninot, Jean-Pierre
WCO	World Customs Organization	A	
WHO	World Health Organization	A	Dr Ustun, Bedirhan
WMO	World Meteorological Organization	A	Mr Shi, Peiliang

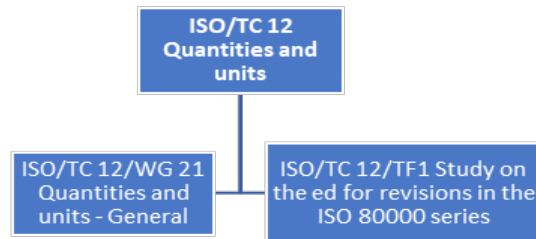
Meetings, held

1978 Copenhagen, Denmark
2010 Prague, Czech Republic
2012 Milan, Italy
2014 Berlin, Germany
2015 Gaithersburg, USA
2016 Stockholm, Sweden
2017 Berlin, Germany
2018-10-08, Finland
2020-05-28, Virtual via Zoom

Meeting, planned

2021-05-26, Virtual via Zoom
2022 (Date: TBD) Onsite, Russian Federation

Committee structure



ISO/TC 12 has one working group and one Task Force directly under the TC:

- **WG 21** “Quantities and units – General”, “*Revision of ISO 80000-1 “Quantities and units— Part 1: General”*”.
Convenor: Prof. Pendrill, Leslie

Meeting held:

2020-02-27 via Zoom
2020-05-27, via Zoom
2021-01-21, via Zoom
2021-02-19, via Zoom
2021-03-19, via Zoom

- **TF1** “*Study on the need for revisions in the ISO 80000 series*”
Convenor: Prof. Pendrill, Leslie

Resolution 137(Zoom 8, 2020)

Other issues arising during 80000 revision ISO/TC 12 decides to create a task force to investigate the need for revision of ISO 80000 standards due to FDIS comments and comments received on the published standards. The members will be, Dr. Ehrlich, Prof. Grimvall, Dr. Knopf, Prof. Obdrzalek, Mr. Pellegrino and Prof. Pendrill (convenor). Liaison representatives will be invited. The Task Force will report to ISO/TC 12 by 2021-03-31.

Meeting held:

2021-01-19, via Zoom
2021-03-26, via Zoom

Business plan

Current ISO/TC 12 Business plan was unanimously approved by ISO/TC 12 at the 2017 meeting in Berlin, Germany (Resolution 05/2017).
[STRATEGIC BUSINESS PLAN ISO/TC 12](#)

Programme of work and published documents

The standards produced by ISO/TC 12 are listed in Annex A.
[Work programme on ISO homepage](#)

Report and resolutions of the last ISO/TC 12 meeting

The resolutions taken and approved at the meeting are to be found in document [ISO/TC 12 N1368](#).

Follow-up on ISO/TC 12 Resolution 130-139 taken via Zoom meeting 2020-05-28

Resolution 130 (Zoom 1, 2020) **OK**

Agenda item 5, Appointment of the drafting committee

ISO/TC 12 decides to appoint Mr. Elster, Mr. Huebner and Mr. Radhe to form the resolutions drafting committee.

The decision was unanimous

Resolution 131 (Zoom 2, 2020) **OK**

Agenda item 7, Business Plan

ISO/TC 12 decides to update the draft Business Plan, N 1353 according to the discussions in the meeting, and tasks the secretary to send the document for an 8 week Committee Internal Ballot (CIB).

The decision was unanimous

Resolution 132 (Zoom 3, 2020) **OK**

Agenda item 8 Report of ISO/TC 12 /WG 21 – General

ISO/TC 12 decides to prepare an 8-week Committee Internal Ballot (CIB) to ask the member bodies if they agree to replace the definitions in clause 3 of ISO 80000-1 by a reference to ISO/IEC Guide 99 (VIM).

It is noted that the definitions in ISO/IEC Guide 99 in ISO 80000-1 only applies to terms used in ISO 80000-1.

The decision was unanimous

Resolution 133 (Zoom 4, 2020) **OK**

Agenda item 8 Report of ISO/TC 12 /WG 21 – General

ISO/TC 12 decides to prepare an 8-week Committee Internal Ballot (CIB) to ask the member bodies if they agree to replace texts about SI in clause 6 of ISO 80000-1 with a general reference to the SI brochure.

It is noted that the SI Brochure will be a normative reference in ISO 80000-1 and thus shall fulfill the requirements in ISO/IEC Directives Part 2, clause 10.2.

The decision was unanimous

Resolution 134 (Zoom 5, 2020) **OK**

Agenda item 8 Report of ISO/TC 12 /WG 21 – General

ISO/TC 12 decides to prepare an 8-week Committee Internal Ballot (CIB) to ask the member bodies if they agree to replace texts in annex A with a general reference to CODATA of ISO 80000-1 by a reference to CODATA.

It is noted that CODATA will be a normative reference in ISO 80000-1 and thus shall fulfil the requirements in ISO/IEC Directives Part 2, clause 10.2.

The decision was unanimous

Resolution 135 (Zoom 6, 2020) **OK**

Agenda item 8 Report of ISO/TC 12 /WG 21 – General

ISO/TC 12 decides to prepare an 8-week Committee Internal Ballot (CIB) to ask the member bodies if they agree to collect the rules regarding printing and writing in ISO 80000-1.

It is noted that there are also some rules in ISO 80000-2 which together with other material should be considered. IEC/TC 25 has an approved work item for printing and writing rules; IEC 80000-16. ISO/TC12 will inform IEC/TC 25 about this decision.

The decision was unanimous

Resolution 136 (Zoom 7, 2020) **Ongoing**

Agenda item 8 Report of ISO/TC 12 /WG 21 – General

ISO/TC 12 decides to prepare a third CD once all the document has updated according to the outcome of the Committee Internal Ballot (CIB).

The decision was unanimous

Resolution 137 (Zoom 8, 2020) **OK, TF 1**

Agenda item 13.6 Other issues arising during 80000 revision

ISO/TC 12 decides to create a task force to investigate the need for revision of ISO 80000 standards due to FDIS comments and comments received on the published standards. The members will be, Dr. Ehrlich, Prof. Grimvall, Dr. Knopf, Prof. Obdrzalek, Mr. Pellegrino and Prof. Pendrill (convenor). Liaison representatives will be invited. The Task Force will report to ISO/TC 12 by 2021-03-31.

The decision was unanimous

Resolution 138 (Zoom 9, 2020) **OK**

Agenda item 13.7 Input for revision of ISO/IEC Directives, Part 2: Principles and rules for the structure and drafting of ISO and IEC documents

ISO/TC 12 decides to form a task force to deliver proposals for texts for ISO/IEC Directives Part 2: Principles and rules for the structuring of ISO and IEC documents regarding rules for quantities and units. To report to ISO/CS by 2020-08-31.

The members will be Mr. Radhe (convenor), Mr. Pellegrino and Prof. Obdrzalek.

The decision was unanimous

Resolution 139 (Zoom 10, 2020) OK

Agenda item 16 Requirements concerning a subsequent meeting

ISO/TC 12 decides to accept the offer from GOST to host the next meeting of ISO/TC 12 in Saint Petersburg in May 2021-05-25 and 26. The committee thanks Russian Member Body for the kind invitation.

The decision was unanimous

Follow up on Resolutions by correspondence 2020:

- Draft Resolution 140, 11/2020 Strategic Business Plan
Vote: 2020-07-18—2020-09-12 **Approved**
- Draft Resolution 141, 12/2020 - Changes to ISO 80000-1
2020-07-23—2020-09-21 **Approved** Vote:
- Draft Resolution 142, 13/2020 - Changes to ISO 80000-1
2020-07-28—2020-09-21 **Approved** Vote:
- Draft Resolution 143, 14/2020 - Changes to ISO 80000-1
2020-07-28—2020-09-21 **Approved** Vote:
- Draft Resolution 144, 15/2020 - Changes to ISO 80000-1
2020-07-28—2020-09-21 **Approved** Vote:

ANNEX A

Standard and/or project under the direct responsibility of ISO/TC 12 Secretariat (5)

[International harmonized stage codes.](#)

[14 Published ISO standards](#)

STANDARD AND/OR PROJECT UNDER THE DIRECT RESPONSIBILITY OF ISO/TC 12 SECRETARIAT (5) ↓	STAGE	ICS
⊙ ISO/CD 80000-1 Quantities and units — Part 1: General	30.92	01.060
⊙ IEC/DIS 80000-6 Quantities and units — Part 6: Electromagnetism	40.99	01.060
⊙ IEC/WD 80000-15 Quantities and units — Part 15: Logarithmic and related quantities	20.20	
⊙ IEC/CD 80000-16 Quantities and units — Part 16: Printing and writing rules	30.00	
⊙ IEC/CD 80000-17 Quantities and units — Part 17: Time dependency	30.00	

Vote in parallel with IEC/TC 25:

***NOTE: IEC 8000-6** “Quantities and units – Part 6: Electromagnetics” is currently under revision in:
IEC/TC 25/JWG 2 - Revision and amendment of IEC-related parts of ISO/IEC 80000 series. Project leader: Prof Göran Grimvall

*NOTE: **IEC 8000-15** “Quantities and units — Part 15: Logarithmic and related quantities” is currently under revision in:

IEC/TC 25/JWG 2 - Revision and amendment of IEC-related parts of ISO/IEC 80000 series.

Project leader: Prof. Jan Obdrzalek.

*NOTE: **IEC 8000-16** “Quantities and units — Part 16: Printing and writing rules” is currently under revision in:

IEC/TC 25/JWG 2 - Revision and amendment of IEC-related parts of ISO/IEC 80000 series. Project

leader: Prof. Dr. Erik Jacobson.

*NOTE: **IEC 8000-17** “Quantities and units — Part 17: Time dependency” is currently under revision in:

IEC/TC 25/JWG 2 - Revision and amendment of IEC-related parts of ISO/IEC 80000 series. Project

leader: Prof. Dr. Erik Jacobson.

ANNEX B

P-members attending ISO/TC 12 meetings 2017-2020

Attending **Not attending**

P-member	2017 Berlin	2018 Finland	2020 Virtual
• Austria (ASI)			
• Belarus (BELST)			
• China (SAC)			
• Czech Republic (UNMZ)			
• Egypt (EOS)			
• Finland (SFS)			
• France (AFNOR)			
• Germany (DIN)			
• Italy (UNI)			
• Japan (JISC)			
• Portugal (IPQ)			
• Russian Federation			

(GOST R)			
• Spain (UNE)			
• Sri Lanka (SLSI)			
• Sweden (SIS)			
• Switzerland (SNV)			
• United Kingdom (BSI)			
• United States (ANSI)			

ISO/TC 12 Secretariat, Stockholm 2021-02-22

Prof Leslie Pendrill (Chair), Mr. Sven Radhe (Committee Manager), Ms. Anette Lindor Norén (Secretariat support) and Katarina Widström (Secretariat support)